

CTDI: CNN-Transformer-Based Spatial-Temporal Missing Air Pollution Data Imputation

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Abstract—Accurate and comprehensive air pollution data is essential for understanding and addressing environmental challenges. Missing data can impair accurate analysis and decision-making. This study presents a novel approach, named CNN-Transformer-based Spatial-Temporal Data Imputation (CTDI), for imputing missing air pollution data. Data pre-processing incorporates observed air pollution data and related urban data to produce 24-hour period tensors as input samples. 1-by-1 CNN layers capture the interaction between different types of input data. Deep learning transformer architecture is employed in a spatial-temporal (S-T) transformer module to capture long-range dependencies and extract complex relationships in both spatial and temporal dimensions. Hong Kong air pollution data is statistically analyzed and used to evaluate CTDI in its recovery of generated and actual patterns of missing data. Experimental results show that CTDI consistently outperforms existing imputation methods across all evaluated scenarios, including cases with higher rates of missing data, thereby demonstrating its robustness and effectiveness in enhancing air quality monitoring. Additionally, ablation experiments reveal that each component significantly contributes to the model's performance, with the temporal transformer proving particularly crucial under varying rates of missing data.

Index Terms—Air pollution, deep learning, transformer, missing data, data imputation, data recovery, spatial-temporal.

I. INTRODUCTION

AIR pollution significantly impacts human health and the environment, especially in urban areas with high levels of pollutants like particulate matter and nitrogen dioxide [1]. These pollutants contribute to serious health issues, particularly affecting vulnerable groups such as children and the elderly [2], leading to millions of premature deaths annually, as estimated by the WHO [3]. Addressing this critical issue requires global initiatives focused on stricter emissions standards, clean energy promotion, and sustainable transportation [4], [5]. High-quality

air quality data is essential for policymakers to identify pollution hotspots [6] and inform targeted strategies [7], [8], and it also enables individuals to make informed health choices [9], [10].

However, obtaining accurate and timely urban air quality data poses challenges [11]. Monitoring stations often suffer from missing data due to equipment malfunctions, sensor failures, power outages, and other issues [12], [13]. Additionally, the high costs of deploying and maintaining monitoring stations limit their coverage in urban areas [14]. To better assess city-wide air pollution, incorporating mobile sensors [15], [16] or developing fine-grained pollution estimation systems [17] is necessary. Yet, low-cost mobile sensors often lack precision, making accurate data estimation even more crucial. Inappropriate recovery of missing data can lead to biased analyses and ineffective interventions, highlighting the need for innovative recovery methods [18].

Recent advancements in machine learning [12], [19], [20] and artificial intelligence [21], [22] present promising solutions for accurately recovering missing air pollution data. The transformer architecture [23], initially designed for natural language processing [24], [25], has proven effective in various fields, including computer vision [26] and time series analysis [27]. This architecture utilizes a Self-Attention mechanism [28], enabling it to focus on relevant input portions and capture long-range dependencies, while also supporting parallelization for handling large datasets [29].

In this study, we propose a Transformer-based data imputation approach for recovering missing air pollution data in urban settings. Our evaluation of this method using real-world data from Hong Kong reveals significant improvements in modeling spatial-temporal (S-T) correlated air pollution. The main contributions of this work are summarized as follows:

- 1) *Innovative CNN-Transformer Architecture*: We propose a groundbreaking approach for recovering air pollution data, termed CNN-Transformer-based Spatial-Temporal Data Imputation (CTDI). This novel architecture synergistically integrates Convolutional Neural Networks (CNNs) and Transformer models, enabling the effective capture of intricate interactions between diverse data types while automatically extracting spatial-temporal (S-T) features. This dual capability enhances the model's performance in accurately reconstructing missing data.
- 2) *Advanced Data Pre-processing Techniques*: The new data pre-processing strategy includes 1) the use of Inverse Distance Weighting (IDW) for spatial interpolation, allowing

Received 7 March 2024; revised 13 January 2025; accepted 20 January 2025. Date of publication 27 January 2025; date of current version 4 September 2025. This work was supported by the Theme-based Research Scheme of the Research Grants Council of Hong Kong SAR Government under Grant T41-709/17-N. Recommended for acceptance by P. D'Urso. (Victor O. K. Li and Jacqueline C. K. Lam contributed equally to this work.) (Corresponding authors: Victor O. K. Li; Jacqueline C. K. Lam.)

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TBDATA.2025.3533882>, provided by the authors.

Digital Object Identifier 10.1109/TBDATA.2025.3533882

for the integration of local urban data—such as meteorological conditions, wind speed, and traffic information—into the air pollution recovery model, and 2) the application of sliding windows (SW) for temporal segmentation, which divides the data into 24-hour period tensors with a 1-hour sliding stride to enhance the input volume.

- 3) *Multi-Data Type Recovery*: Unlike conventional data imputation methods that are limited to handling a single type of data, CTDI can simultaneously recover multiple types of S-T air pollution data. This is achieved through the incorporation of 1-by-1 CNN layers that capture interactions between different data types. By preserving the relationships between different pollutants, rather than discarding or overlooking this information, the recovery performance of the model is improved.
- 4) *Pioneering S-T Transformer Module*: The S-T transformer module is a key innovation in our approach, consisting of both spatial and temporal transformers to extract relevant features in both dimensions. This represents the first application of transformer architecture specifically for the recovery of missing air pollution data, marking a significant advancement in the field.
- 5) *Real-World Dataset Analysis*: We conduct a comprehensive analysis of missing data from a real-world air pollution dataset in Hong Kong, offering valuable insights into the nature and patterns of data losses. This empirical evaluation not only demonstrates the effectiveness of our model but also provides a robust foundation for further exploration and refinement of air quality data recovery techniques.

The organization of this paper is as follows. Section II reviews existing data imputation approaches and motivates the use of the transformer architecture in the recovery of air pollution data. In Section III, the proposed CTDI model is introduced in detail. Section IV follows with a statistical analysis of real-world missing air pollution data in Hong Kong, an examination of the performance of CTDI, and a discussion of the results. Finally, the paper concludes in Section V and offers suggestions for future research directions.

II. RELATED WORK

Existing data imputation approaches can be broadly classified into three categories: 1) Statistical imputation [30], [31], 2) Machine learning-based imputation [32], [33], [34], and 3) Deep learning-based imputation [35], [36], [37].

Statistical imputation methods primarily use statistical techniques to estimate missing values based on the observed data. A well-known approach within this category is Multiple Imputation [38], [39]. Instead of filling in missing values with a single estimate (such as the mean), multiple imputation creates several different plausible datasets by imputing the missing values multiple times [40]. In contrast, shallow machine learning-based imputation leverages traditional machine learning algorithms, such as decision trees or k-nearest neighbors, to predict and fill in missing values [32]. These methods typically use patterns

in the available data to infer the missing entries, optimizing for accuracy and efficiency. While they can be effective, they often rely on well-defined feature relationships and may struggle with highly complex datasets compared to their deep-learning counterparts.

A. Deep Learning Based Approaches to S-T Data Imputation

Deep learning has been fruitfully applied across various fields [41], [42] and offers several advantages over statistical and shallow machine learning techniques [43]. First, deep learning algorithms automatically learn features from raw data, minimizing the need for manual feature engineering common in statistical imputation. Second, they capture intricate patterns, allowing for better handling of nonlinear relationships and high-dimensional feature spaces than traditional methods. Third, deep learning models exhibit excellent scalability, robustness, and transferability. Traditional deep learning based S-T data imputation approaches can be categorized into three types: 1) Convolutional Neural Network (CNN)-based models [44], [45], 2) Recurrent Neural Network (RNN)-based models [46], [47], and 3) CNN-RNN Hybrid models [48], [49].

Convolutional Neural Networks (CNNs) are classical deep learning models, initially designed for processing and analyzing visual data, such as images and videos [50], [51], [52], [53]. Typical CNN-based data imputation methods include Autoencoder-based Multiple Imputation using Denoising Autoencoders (MIDA) [35] and Graph-based Imputation Neural Networks (GINN) [44]. MIDA imputes missing values by training a CNN-based autoencoder to learn the underlying patterns of complete data and then uses this knowledge to generate plausible imputations. GINN utilizes graph convolutional neural networks to capture the complex relationships between nodes and imputes missing values based on the surrounding graph structure. However, CNN-based methods struggle with long-range dependencies in spatial-temporal (S-T) data and are computationally intensive, limiting their practicality for imputing S-T correlated data.

Recurrent Neural Networks (RNNs) are designed for processing sequential data, such as time series [54], text [55], and speech [56]. They maintain a hidden state as memory, allowing them to capture temporal dependencies and contextual information. Long Short-Term Memory (LSTM) networks [57], [58] address the vanishing gradient problem, enabling the modeling of long-term dependencies. Gated Recurrent Units (GRUs) [59] simplify LSTMs with similar gating mechanisms. Despite their effectiveness, RNNs have limitations with S-T data [60], [61]: 1) they struggle to capture spatial dependencies, 2) they process data in a sequential manner, which limits their ability to utilize parallel processing, and 3) they face memory constraints that hinder the retention of long-term dependencies.

CNN-RNN hybrid models combine the strengths of CNNs and RNNs for handling S-T data [62], [63]. CNNs excel at extracting spatial features, while RNNs capture temporal dependencies. However, hybrid models are more complex, with

a larger number of parameters that increase the computational intensity and training time [64].

B. Transformer

The transformer deep learning architecture was proposed by Vaswani et al. in the 2017 paper "Attention Is All You Need" [65]. It is able to capture long-range dependencies and achieve state-of-the-art results in tasks such as machine translation, text generation, sentiment analysis, and more [24], [26], [66]. Unlike traditional sequential models, such as recurrent neural networks (RNNs) or convolutional neural networks (CNNs), the transformer relies on a Self-Attention mechanism [67] to process input sequences. The Self-Attention mechanism allows the model to weigh the importance of different positions within the input sequence, enabling it to focus on relevant information and disregard irrelevant or redundant parts. This gives the transformer its power and flexibility and, by resolving the issues of vanishing and exploding gradients associated with RNNs, also greatly improves the modelling of long-range dependencies.

Furthermore, transformer models can be trained in parallel, making them much quicker to train than sequential models. This parallelization results from the Self-Attention mechanism's ability to process input sequences in parallel, eliminating the sequential bottleneck present in traditional models. Additionally, Multi-Head Attention [68] allows the transformer model to jointly attend to information from different representation subspaces at different positions. This enhanced expressivity helps the model extract more subtle and complex feature patterns in the data.

The transformer architecture has been widely adopted and extended since its introduction. Variants such as BERT (Bidirectional Encoder Representations from Transformers) [69], GPT (Generative Pre-trained Transformer) [70], and T5 (Text-to-Text Transfer Transformer) [71] have achieved remarkable results on a wide range of NLP tasks. The Transformer's impact extends beyond NLP, as researchers have also applied it to computer vision, speech recognition, and other domains. Regarding S-T data studies, Mingxing Xu et al. have successfully combined GCN and Transformer structure to forecast traffic flow [72]. Weibin Zhang et al. have incorporated Self-Attention into a Generative Adversarial Network (GAN) to repair missing data for traffic flow [73]. Qiaolin Zeng et al. have combined a data decomposition strategy with transformer network to predict the PM_{2.5} time-series [74]. Given the exceptional performance demonstrated by the transformer in these studies, we have compelling evidence to support its efficacy as a potent solution for addressing missing air pollution data.

III. METHODOLOGY

In this study, we propose CTDI (CNN-Transformer-based Data Imputation), which is a deep learning framework that combines CNN and transformer architectures to recover missing air pollution measurements. This is shown schematically in Fig. 1. CTDI consists of three components: a data pre-processing component, a 1×1 convolutional component, and an S-T Transformer module. The remainder of this section describes

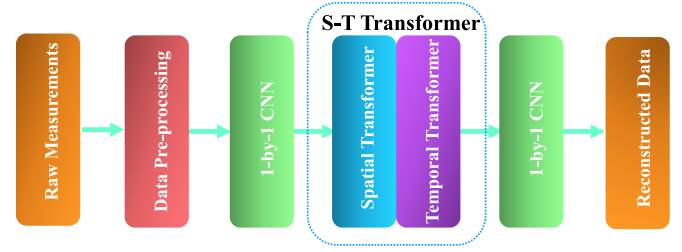


Fig. 1. Overall framework of CTDI.

the design of each component in detail followed by the loss function.

A. Data Pre-Processing Component

Certain urban factors, such as wind speed and traffic information, may not be in the immediate vicinity of the air quality stations yet still affect air pollution. To incorporate these features into the CTDI framework, the data pre-processing component employs a spatial interpolation mechanism to map these features to each air pollution monitoring station. Specifically, Inverse Distance Weighting (IDW) is used with a squared weighting function [75]. The formula is as follows:

$$u'_j = \begin{cases} \frac{\sum_{i=1}^N w_{ij} u_i}{\sum_{i=1}^N w_{ij}} & d(i, j) \neq 0 \\ u_i & d(i, j) = 0 \end{cases}, \quad (1)$$

where u_i denotes the raw urban data at the i location, and u'_j denotes the mapped urban data at the j location at which an air quality monitoring station is situated. $w_{ij} = \frac{1}{d(i, j)^p}$ is the inverse distance weight of the data and $p = 2$. By assigning higher weights to nearby locations and lower weights to more distant locations, the squared weighting function increases the effect of urban features closer to a particular monitoring station.

After spatial interpolation, the dataset available for imputation contains air pollution data and mapped urban data. This can be regarded as a large 3-dimensional tensor. Let $\mathbf{X}_{raw} \in \mathbb{R}^{s \times t \times c}$ denote this raw data tensor, where s denotes spatial features (i.e. individual monitoring stations); t denotes the temporal features (i.e. time of measurement); and c represents the number of data categories, and is comprised of $c = c_1 + c_2$, i.e., c_1 kinds of air pollutants and A_{c_2} kinds of urban factors.

The raw data tensor is then sliced into multiple 24-hour tensor samples, which will form the inputs for data recovery by the model. This temporal segmentation is done using a sliding window technique similar to that used in RNN-based prediction models [76], in which a data point is predicted based on a finite set of previous data points within the dataset. As depicted in Fig. 2, a sliding window of fixed size 24 in the time dimension is used to extract the 24-hour tensor $\mathbf{X} \in \mathbb{R}^{s \times t_d \times c}$, where $t_d = 24$ (i.e., one 24-hour period). A sliding stride of 1 h then moves the window in 1-hour increments to generate new 24-hour tensors. Utilizing sliding windows along the time axis increases the number of 24-hour tensor samples, which increases the size of the training dataset. As stability and potentially also the accuracy of deep learning methods improve as the size of the training dataset increases [37], this improves the performance of CTDI.

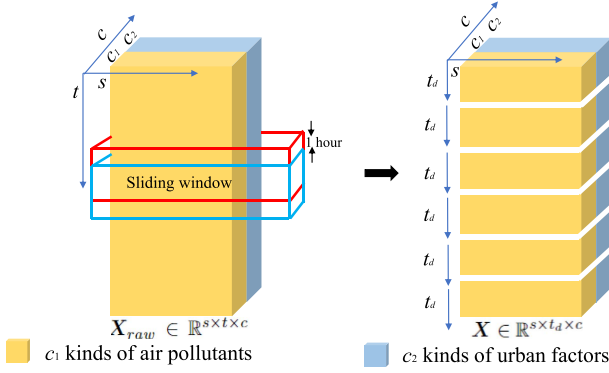


Fig. 2. Sliding window for temporal segmentation.

Additionally, some entries in the 24-hour data tensor $\mathbf{X}$ are erased to imitate missing air pollution data. This is done using a mask. Let $\mathbf{M} \in \mathbb{R}^{s \times t_d \times c}$ be the binary mask tensor, corresponding to the distribution of the missing entries in the 24-hour tensor sample. The formal definition of $\mathbf{M}$ is as follows:

$$\mathbf{M}(i, j, k) = \begin{cases} 1 & \mathbf{X}(i, j, k) \text{ is observed} \\ 0 & \mathbf{X}(i, j, k) \text{ is missing} \end{cases}, \quad (2)$$

where (i, j, k) represents the index of individual entry in $\mathbf{X}$ or $\mathbf{M}$. As our focus is on recovering missing air pollution data, the missing entries are restricted to air pollution features. Let $\odot$ denote entry-wise multiplication. Then $\mathbf{X} \odot \mathbf{M}$ is a 24-hour tensor sample that is incomplete (in the sense that it is missing air pollution data) and has had each missing data entry assigned as 0.

B. 1-By-1 CNN Layers

After data pre-processing, a single 1-by-1 CNN layer is introduced before and after the transformer module to capture interactions between different data types. 1-by-1 convolutions, also known as point-wise convolutions, make the model richer by enabling non-linear feature transformations and control the size of the tensor by adjusting the number of channels [77], [78].

In the tensor $\mathbf{X} \in \mathbb{R}^{s \times t_d \times c}$, different types of data (such as $\text{PM}_{2.5}$ measurements and temperature) are each allocated to a specific channel within the tensor. The utilization of 1×1 convolutions followed by the application of non-linear activation functions, such as ReLU (Rectified Linear Unit), after the 1×1 convolutions allow the model to learn complex interactions between input features. Importantly, this includes interactions between different types of data since information from different channels is combined, so the model is able to better attend to relevant features and to generate more expressive representations. This non-linear feature transformation significantly enhances the network's modelling flexibility.

As shown in Fig. 3, by utilizing f 1-by-1 convolution filters of size $1 \times 1 \times c$, the number of channels in the incomplete 24-hour data tensor is transformed from c to f . After the transformer module, c 1-by-1 convolution filters are employed to construct an output data tensor with the same dimensions as the input 24-hour data tensor $\mathbf{X} \in \mathbb{R}^{s \times t_d \times c}$, as depicted in Fig. 4. As

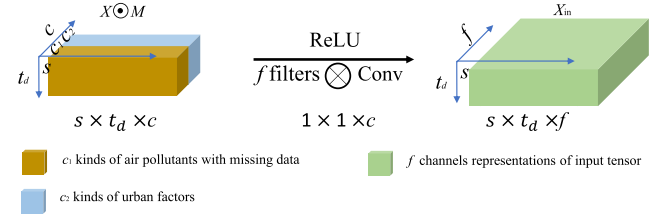


Fig. 3. 1-by-1 CNN Layer before Transformer module.

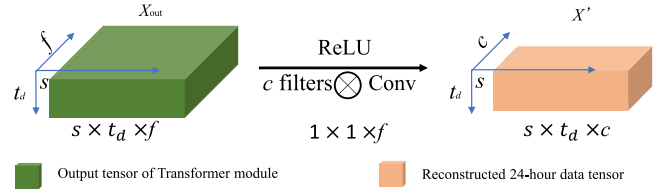


Fig. 4. 1-by-1 CNN Layer after Transformer module.

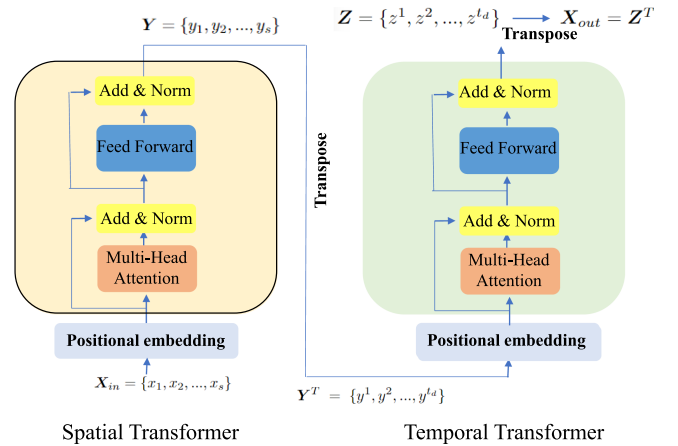


Fig. 5. S-T Transformer Module.

1-by-1 CNN layers conserve the spatial and temporal shape of the input tensor, positional relationships are preserved.

C. S-T Transformer Module

The S-T transformer module aims to capture the spatial and temporal relationships of the input tensor $\mathbf{X}_{in} \in \mathbb{R}^{s \times t_d \times f}$ generated by the 1×1 CNN layer. As shown in Fig. 5, this module consists of two Transformer units, namely one spatial transformer and one temporal transformer. Each transformer module is comprised of the following components: 1) one Positional Embedding layer, which assigns unique embeddings to each position in the input sequence. 2) one Multi-Head Attention layer, which processes multiple Self-Attention calculations (scaled dot-product attention) in parallel to extract features from the input sequence. 3) two Add & Norm layers, in which the Add step comprises a residual connection to address the vanishing gradient problem and the Norm step comprises layer normalization. 4) one Feed Forward layer, which is a position-wise transformation to enhance the feature extraction by introducing non-linearity.

We now describe how an input sequence is processed. The input data tensor $\mathbf{X}_{in} \in \mathbb{R}^{s \times t_d \times f}$ can be regarded as a set of time-series data from s stations and denoted as $\mathbf{X}_{in} = \{x_1, x_2, \dots, x_s\}$, where $x_i \in \mathbb{R}^{t_d \times f}$ represents a 24-hour period of data from the i th station. The spatial transformer transforms this station based sequence into $\mathbf{Y} = \{y_1, y_2, \dots, y_s\}$, where $y_i \in \mathbb{R}^{t_d \times f}$ and $\mathbf{Y} \in \mathbb{R}^{s \times t_d \times f}$. Next, $\mathbf{Y} \in \mathbb{R}^{s \times t_d \times f}$ is transposed to $\mathbf{Y}^T \in \mathbb{R}^{t_d \times s \times f}$, where the sequence $\mathbf{Y}^T$ can be denoted as $\mathbf{Y}^T = \{y^1, y^2, \dots, y^{t_d}\}$ and $y^j \in \mathbb{R}^{s \times f}$ represents the air quality-related data for all s stations in the j th hour of a given 24-hour period. The temporal transformer then outputs $\mathbf{Z} = \{z^1, z^2, \dots, z^{t_d}\}$, where $z^j \in \mathbb{R}^{s \times f}$ and $\mathbf{Z} \in \mathbb{R}^{t_d \times s \times f}$. Finally, the output data tensor $\mathbf{Z}$ is transposed to give $\mathbf{X}_{out} \in \mathbb{R}^{s \times t_d \times f}$, i.e., $\mathbf{X}_{out} = \mathbf{Z}^T$.

In the original transformer model, Vaswani et al. [65] apply positional encoding to represent the order of the sequence in the positional embedding layer and utilize Multi-Head Attention to capture different types of dependencies and to attend to relevant information across various positions of the input sequence simultaneously. The proposed CTDI follows Vaswani et al. in the configuration of positional encoding and Multi-Head Attention in spatial and temporal transformers. The specific configurations are detailed in Appendix A, available online.

Additionally, each component of the CTDI model, described in Sections III-B and III-C, has been thoroughly analyzed in the ablation study detailed in Appendix B, available online. This comprehensive examination reveals the distinct contributions and impacts of each component on the overall performance of the proposed model, providing critical insights into their significance and effectiveness.

D. Loss Function

We now turn to discuss the loss function of this model. As defined in previous sections, $\mathbf{X} \in \mathbb{R}^{s \times t_d \times c}$ represents the ground truth 24-hour data tensor. $\mathbf{M} \in \mathbb{R}^{s \times t_d \times c}$ is the corresponding binary mask tensor, in which the observed entry is assigned 1 and the missing entry is 0. Let $C(\cdot)$ denote the mathematical formulation of our CTDI network. Given an incomplete matrix $\mathbf{M} \odot \mathbf{X}$ as the input, the network generates an estimated 24-hour data tensor $\mathbf{X}' \in \mathbb{R}^{s \times t_d \times c}$ in the form of $C(\mathbf{M} \odot \mathbf{X})$.

To improve the accuracy of the air pollution data recovery, a coupled loss function is incorporated into the CTDI model. The first term is the Imputation Loss, L_{im} , which captures the difference between the ground truth and the estimated values of the missing entries in the 24-hour data tensor. Mathematically, this is expressed in (3) below:

$$L_{im}(\mathbf{X}, \mathbf{M}) = \|(1 - \mathbf{M}) \odot (\mathbf{X} - C(\mathbf{M} \odot \mathbf{X}))\|_2^2 \quad (3)$$

$$= \|(1 - \mathbf{M}) \odot (\mathbf{X} - \mathbf{X}')\|_2^2, \quad (4)$$

where $(1 - \mathbf{M})$ makes the missing entries equal to 1 and $\|\cdot\|_2^2$ represents the square of the euclidean distance of the missing entries in the tensor. However, L_{im} reflects the average imputation error of all missing entries but not the entry-wise imputation precision, which also affects the validity of the reconstructed matrix. Accordingly, a further loss term, L_{re} Reconstruction

Loss, is introduced:

$$L_{re}(\mathbf{X}, \mathbf{M}) = \|\mathbf{M} \odot ((\mathbf{X} - C(\mathbf{M} \odot \mathbf{X})))\|_2^2 \quad (5)$$

$$= \|\mathbf{M} \odot (\mathbf{X} - \mathbf{X}')\|_2^2, \quad (6)$$

where $\mathbf{M}$ ensures all the observed entries are assigned to 1. Equation (5) represents the differences between the real observed values and the reconstructed values. By minimizing this term, the accuracy of individual observations in the reconstructed 24-hour data tensor can be preserved. Similar to the accumulated loss used in [79], the Reconstruction Loss can improve the accuracy of the reconstructed matrix and convergence speed. However, in practice, the loss function L_{re} is much larger than L_{im} , so to avoid L_{re} dominating the direction of the loss function's gradient descent, a scaling factor is used to adjust the relative size of these two loss terms. Through trial and error, the CTDI model was calibrated to $\alpha = 1000$ and $\beta = 1$.

$$L(\mathbf{X}, \mathbf{M}) = \alpha L_{im}(\mathbf{X}, \mathbf{M}) + \beta L_{re}(\mathbf{X}, \mathbf{M}) \quad (7)$$

By minimizing $L(\mathbf{X}, \mathbf{M})$, the CTDI model can be trained for accurate missing data recovery.

IV. EXPERIMENTAL SET-UP

To evaluate the recovery performance of the proposed CTDI model, a series of simulations were conducted. This section begins by introducing the dataset used and analyzing its patterns of missing data. Next, performance metrics for the proposed model and comparison baselines are presented. Finally, simulation results for different cases of missing data are evaluated and discussed.

A. Datasets

The dataset used was obtained from three online Hong Kong databases, namely the Hong Kong Environmental Protection Agency (HKEPD) Database [80], Hong Kong Observatory Open Database [81], and Hong Kong Traffic Speed Map (City Dashboard Version) Database [82]. As shown in Table I, the dataset contains: 1) five types of air pollutants, 2) six types of meteorology data, and 3) two types of traffic data. The number of data nodes¹ and frequency of data update are also shown. The data retrieved was from 2019-01-01 to 2021-12-31. To standardize the frequency of each type of data to one hour (for consistency with the air pollution data), data updated more than once per hour was averaged over the hour.

After spatial interpolation, the size of the original data tensor $s \times t \times c = 16 \times 26304 \times 13$. The original dataset hence consists of 5,471,232 items in total. Next, we used a random 80/10/10 split of the raw dataset as the training set, the validation set, and the test set. The raw data in each set was then divided into multiple 24-hour period data tensor samples, using a temporal sliding window to augment the data.

¹ Air pollution data from two new air pollution monitoring stations in Hong Kong are not included for data consistency.

TABLE I
HONG KONG DATASET COLLECTED FROM 2019-01-01 TO 2021-12-31

Domain	Number of Data Nodes	Data Category	Unit	Update Frequency
Air pollution	16 stations	PM _{2.5}	$\mu\text{g}/\text{m}^3$	1 hour
		PM ₁₀	$\mu\text{g}/\text{m}^3$	
		NO ₂	$\mu\text{g}/\text{m}^3$	
		SO ₂	$\mu\text{g}/\text{m}^3$	
		O ₃	$\mu\text{g}/\text{m}^3$	
Meteorology	47 stations	Pressure	hPa	10 min
		Relative humidity	%	
		Temperature	°C	
		Visibility	km	
		Wind direction	N/A	
Traffic	607 roads	Wind speed	km/h	5min
		Traffic speed	km/h	
		Traffic congestion	N/A	

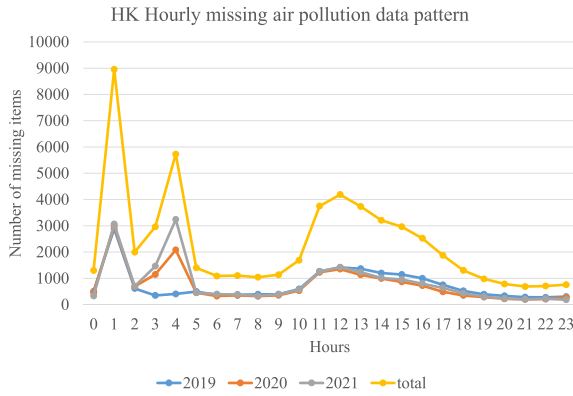


Fig. 6. HK hourly missing air pollution data in different years.

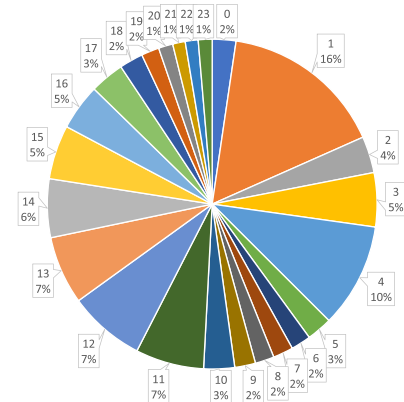


Fig. 7. Proportion of hourly missing data in the total missing data.

B. Missing Data Patterns Analysis and Design

A detailed statistical analysis was first conducted to characterize the missing air pollution data. The air pollution measurements cover 5 types of air pollutants from 16 Hong Kong monitoring stations. There are 55,875 missing data items, which corresponds to an arithmetic mean missing data rate of 2.46%. However, the rates of missing data within any given 24-hour period are diverse and the highest rate of missing data exceeds 50%.

Fig. 6 displays the distribution of missing data by hour in different years. There is some variation in amplitude, notably between 2:00 am and 5:00 am, but the shape of the hourly missing data curves is broadly similar across the different years. Missing data is particularly pronounced at 1:00 am, 4:00 am, and 12:00 pm. Additionally, there is more missing data in the early morning and around noon than in other periods. Fig. 7 shows the proportion of missing data by hour in the total air pollution dataset. One potential reason for this non-uniform distribution may be scheduled downtime for maintenance and inspection. Additionally, it may be more difficult to resolve unscheduled interruptions (for example device faults, network problems, communication interruptions, and power failures) in a timely manner at certain hours.

As depicted in Fig. 8, the proportion of missing data by air pollutant is much more uniformly distributed. Looking at

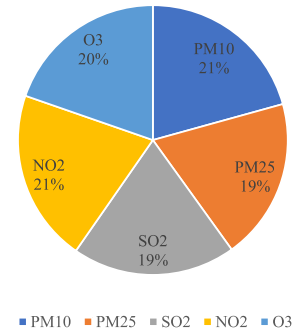


Fig. 8. Proportion of different pollutants in the total missing data.

missing data rates for each pollutant through time, NO₂, O₃, and SO₂ have similarly shaped distributions that occur at the same times, as do PM_{2.5} and PM₁₀. We hypothesize that this may be caused by a single device measuring different air pollutants, so a problem with the device affects the measurements of all those pollutants simultaneously. Fig. 9 shows the proportion of missing data at different monitoring stations and lies between missing data by hour and by air pollutant in terms of uniformity.

Following the above investigation, data masks to simulate missing data were created. In the first four cases, the entries that were erased were randomly chosen, subject to case-specific

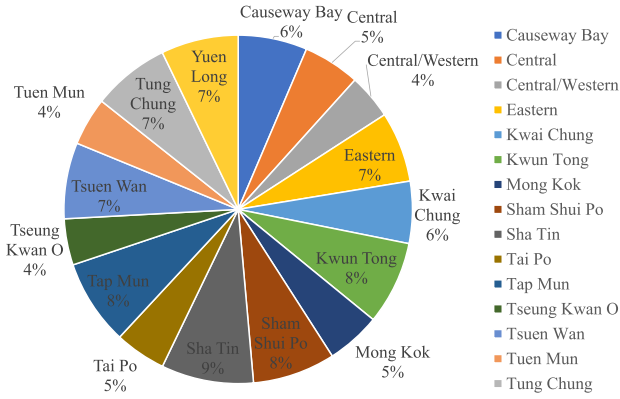


Fig. 9. Proportion of missing data from different stations in the total missing data.

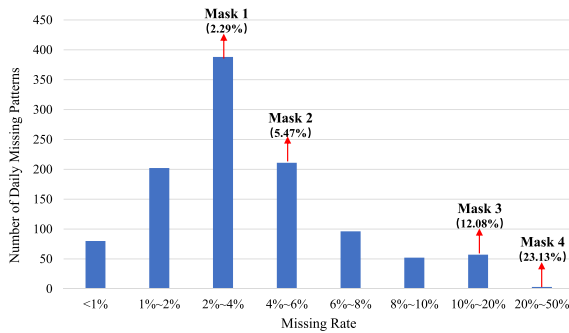


Fig. 10. Four missing data mask patterns with different missing rates sampled from real-life daily missing data from 2016 to 2018.

constraints. Multiple masks were generated for each missing data rate.

We began by investigating the general performance of the model across a range of missing data rates:

Case 1: Low missing data rates of 2%, 4%, 6%, 8%, and 10% within any given 24-hour period.

Case 2: High missing data rates of 20%, 30%, 40%, 50%, and 60% within any given 24-hour period.

Next, we proceeded to examine the performance of the model in recovering missing data calibrated to our real-world dataset:

Case 3: Missing data rate of 2.46% across the total dataset (i.e. same average missing data rate as real-world total dataset). Additionally, the proportions of missing data by hour, pollutant, and monitoring station were controlled to be as characterized in the pie charts.

Case 4: Missing data rate of 24.60% within any given 24-hour period. Additionally, the proportions of missing data by hour, pollutant, and monitoring different station were controlled to be as characterized in the pie charts.

Lastly, we tested the effectiveness of the model in practice by using it to recover missing data for masks that reproduced actual missing data patterns from the real-world dataset. Fig. 10 shows the frequency of different proportions of missing data within a 24-hour period. Further examination showed that the majority of missing items across all patterns are consecutive (as

opposed to discontinuous or sporadic).² *Case 5* comprises four masks sampled from actual data. Mask 1, with a missing data rate of 2.29%, was randomly chosen from the most common interval (2–4%). Mask 2, with a missing data rate of 5.47%, was randomly chosen from the second most common interval (4–6%). Mask 3 and Mask 4, with respective missing data rates of 12.08% and 23.13%, were randomly chosen from the two intervals with the most severe missing data.

C. Simulation Configuration

The proposed CTDI model was evaluated against existing data imputation approaches (see Section II). Nine baseline models were utilized, including 1) Mean Imputation and Multivariate Imputation by Chained Equations (MICE), two traditional statistical imputation approaches [83], [84]; 2) K-nearest neighbors (KNN) and Miss Random Forest (MissForest), which are machine learning-based imputation methods; 3) MIDA, LSTM, and GINN, which represent deep learning-based imputation techniques; and 4) two advanced methods that we previously proposed, namely, Optimization-based Singular Value Thresholding (SVT) [12], specifically designed for short-term data interpolation, and the state-of-the-art GCN-based S-T missing pattern identification and recovery (GCN-ST-MDIR, simplified in this paper as GSTR) method [11], which adaptively assigns the Long-Short Term Context Encoder (LSCE) [37] and Improved Long-Short Term Context Encoder (ILSCE) [13] to recover different missing data patterns. The batch size and learning rate for the deep learning-based baseline models and CTDI were set to 32 and 0.002, respectively. The max epoch for CTDI was set to 100. The optimal epoch of the training process is reached when the validation error takes its optimal value within the maximum epoch.

To evaluate the recovery accuracy of the proposed method, mean absolute percentage error (MAPE) was used as the performance metric. MAPE [17] is a classical statistical metric to assess the accuracy of forecasting or prediction models and is widely used in various fields, such as economics, finance, and supply chain management. For the missing data recovery task, MAPE measured the average percentage deviation between recovered and actual values as follows:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\mathbf{X}_m(i, j, k) - \mathbf{X}'_m(i, j, k)}{\mathbf{X}_m(i, j, k)} \right| \times 100\% \quad (8)$$

where $\mathbf{X}_m(i, j, k)$ is the missing data entry, of which the binary mask value is equal to 0, i.e., $M_m(i, j, k) = 0$. $\mathbf{X}'_m(i, j, k)$ represents its corresponding recovered value generated by CTDI model. n is the total number of missing air pollution data in the incomplete 24-hour tensor $\mathbf{X} \odot \mathbf{M}$.

²Consecutive missing data contains four kinds of missing data: (1) Temporal consecutive: the data are lost for a period of time; (2) Spatial consecutive: the data are lost in different stations simultaneously; (3) Pollutants consecutive: the data of different pollutants are lost simultaneously at the same station in the same period; (4) Mixed consecutive: the most commonly missing case which mixes the first three categories of consecutive missing together. At the same time, sporadic loss represents occasional data loss, which is different from the above consecutive loss.

TABLE II
DATA RECOVERY COMPARISON FOR CASE 1

MAPE (%)										
Missing rate	Mean	MICE	KNN	MissForest	MIDA	LSTM	GINN	SVT	GSTR	CTDI
2%	15.96	15.98	16.30	16.12	15.45	17.97	14.98	14.93	13.23	13.29
4%	18.33	15.49	17.56	17.04	16.39	18.22	16.00	15.52	13.98	13.91
6%	19.21	17.96	18.69	18.88	17.62	18.14	16.42	15.99	13.57	13.59
8%	20.09	19.46	20.50	19.95	19.51	20.82	17.85	17.05	14.05	13.88
10%	23.45	23.22	22.03	23.00	20.93	21.89	18.36	18.63	14.59	14.06

TABLE III
DATA RECOVERY COMPARISON FOR CASE 2

MAPE (%)										
Missing rate	Mean	MICE	KNN	MissForest	MIDA	LSTM	GINN	SVT	GSTR	CTDI
20%	28.14	26.96	25.52	26.24	23.40	22.61	20.04	26.84	14.98	14.45
30%	32.95	27.58	27.98	25.98	24.01	23.16	21.32	29.76	16.19	15.05
40%	37.87	30.37	30.11	29.85	26.85	25.07	23.71	32.33	18.13	15.90
50%	44.20	37.88	37.50	35.72	30.82	28.15	26.99	42.59	21.35	17.26
60%	60.79	45.03	48.08	41.21	35.67	32.01	33.47	56.58	25.94	18.57

V. EXPERIMENTAL RESULTS AND DISCUSSIONS

A. CTDI Performance in Case 1 and Case 2

We first compare the data recovery performance of CTDI against the nine baseline models across a range of missing data rates for Case 1, comprising scenarios with low rates of missing data, and Case 2, comprising scenarios with high rates of missing data.

As illustrated in Tables II and III, recovery errors for all imputation methods increase as the missing data rates rise. Compared with traditional statistical and shallow machine learning-based imputation methods, deep learning-based methods are both more accurate and deteriorate by less as missing data increases. Significant performance differences among the various imputation methods can be seen at different missing data rates.

Specifically, Mean Imputation performs the worst in both Cases, showing a substantial increase in MAPE values as the missing rate rises, reaching 60.79% at a 60% missing rate. MICE shows slightly better performance at lower missing rates but similarly struggles at higher rates. KNN and MissForest exhibit slightly better results but are limited by strict data distribution assumptions, which hinder their effectiveness in spatial-temporal data recovery. Among deep learning methods, MIDA and LSTM show some improvement but still fall short compared to GINN. The GINN model benefits from graph construction, allowing it to consider the relevance of different data nodes, significantly enhancing recovery accuracy. SVT performs well in low-rate missing data interpolation but does not stand out at higher missing rates. GSTR demonstrates good adaptability in handling spatial-temporal missing patterns, showing marked improvement over traditional methods, especially at higher missing rates. The combination of graph-based modelling and adaptive context encoding strategies contributes to GSTR's superior performance in data recovery compared to GINN, MIDA, and LSTM.

Our CTDI model consistently outperforms all other methods across both cases. In Case 1, CTDI achieves a MAPE of only 14.06% at a 10% missing rate, while in Case 2, it reaches a MAPE of 18.57% at a 60% missing rate. The exceptional

performance of CTDI can be attributed to its transformer architecture, which effectively captures sequential dependencies and spatial-temporal relationships, resulting in superior data recovery capabilities. The ability to focus on specific parts of the input data through the attention mechanism allows Transformers to capture complex relationships and interactions that may be missed by CNNs, which focus on local features, or by GCNs, which depend on graph structures. Meanwhile, unlike sequential computations of LSTM and GSTR, CTDI leverages parallel computation, significantly enhancing optimization efficiency, particularly in scenarios with high missing data.

B. CTDI Performance in Case 3-5

Case 3 and Case 4 calibrate the generated missing data masks to the data actually missing from the real-world air pollution dataset observed in Hong Kong. Both cases control the proportion of missing data by hour of day, pollutant, and monitoring station to be the same as the observed dataset. Case 3 additionally constrains the rate of missing data rate to be 2.46% across the total dataset, which corresponds to the overall missing data rate in the full observed 3-year air pollution dataset, while Case 4 constrains the rate of missing data to be 24.60% within any given 24-hour period to test performance under much more severe data loss.

Table IV presents the recovery performance of the CTDI and baseline models. Again, CTDI significantly outperforms the other models. Tables V and VI break down the recovery error of CTDI in Case 3 and Case 4. Table V presents the recovery error of CTDI by air pollutants. It can be seen that the recovery of SO₂ and O₃ data is more difficult than that of NO₂, PM_{2.5}, and PM₁₀, since the recovery errors are higher despite the proportion of missing data by pollutant being nearly uniform. This may be due to varying feature correlations between different types of data. The 1-by-1 CNN layer captures interactions between air pollution data and urban data, comprised of traffic conditions and meteorological data. Traffic conditions may be particularly relevant for NO₂, PM_{2.5}, and PM₁₀ as vehicular emissions and

TABLE IV
DATA RECOVERY COMPARISON FOR CASE 3 AND CASE4

Missing rate	MAPE (%)									
	Mean	MICE	KNN	MissForest	MIDA	LSTM	GINN	SVT	GSTR	CTDI
2.46%	14.35	14.93	15.60	15.33	14.07	18.48	13.88	14.13	12.64	10.90
24.60%	31.12	26.76	26.88	26.12	22.65	21.84	20.70	28.25	14.29	13.05

TABLE V
CTDI ACCURACY OF RECOVERING DIFFERENT AIR POLLUTANTS
FOR CASE 3 AND CASE4

MAPE (%) for Different Air Pollutants					
Missing rate	NO ₂	O ₃	PM _{2.5}	PM ₁₀	SO ₂
2.46%	9.92	12.41	10.38	7.82	13.97
24.60%	11.01	14.60	12.92	10.41	16.29

tire/road wear are key sources of NO_x and urban particulate matter. Meteorological conditions may be more important for O₃ and SO₂, since O₃ is typically formed by photochemical reactions between NO_x and volatile organic compounds (VOCs) in the presence of sunlight and SO₂ is typically generated by the burning of fossil fuels (including by power stations and ships). Table VI presents the recovery performance of CTDI by hour of day. The five hours with the highest MAPE are shown in bold. The recovery accuracy of CTDI decreases as the number of missing entries increases, hence for our real-world dataset in which there is more missing data in the early morning and noon periods, it is more difficult to impute the data that is missing in these periods than other times of the day.

Case 5 evaluates the data recovery performance of CTDI for four masks that each replicate different real-world missing air pollution patterns. The results are shown in Table VII. As in Cases 1–4, the proposed CTDI method consistently outperforms all other baseline approaches. The out-performance of CTDI with real-world data loss and the consistency of CTDI's out-performance across multiple different scenarios strongly suggest that CTDI can be usefully deployed in the real-world and improve the recovery accuracy of missing air pollution data.

C. Test for Different Datasets

The selection of datasets for missing data imputation has been shown in previous work [11], [37] to be important for the performance of deep learning models. In this section, we evaluate the performance of the proposed CTDI method using two different datasets. The first dataset, as previously described, comprises five types of air pollutants, six types of meteorology data, and two types of traffic data. For convenience, we will refer to this dataset as “AU” for air pollution data and urban data. The second dataset includes only the five types of air pollutants and will be referred to as dataset “A”.

Table VIII presents the performance of CTDI for the two datasets. The performance of CTDI for the “AU” dataset and the “A” dataset are similar, but data recovery for the “A” dataset is generally being slightly better than for the “AU” dataset. Although data on traffic and other urban factors may be important for forecasting air pollution, the results indicate that a dataset comprised of only air pollution measurements is sufficient for

CTDI to recover missing data in this Hong Kong case study. We believe that the information contained in multiple types of observed air pollutants before, during, and after the timestamp are enough to reconstruct the loss [11] because: 1) The observed air pollution measurements (including observations from nearby monitoring stations) already incorporate the impact of urban factors. 2) In the Hong Kong dataset, the urban factors occur at locations different to those of the air pollution monitoring stations. While spatial interpolation was employed to align the urban factors with the air pollution measurements from the monitoring stations, the resulting accuracy of the urban data for each station is insufficient to add new information to that already contained in the air pollution data.

However, incorporating urban data does have a positive effect by expanding the tensor size of the 24-hour period input data. Similar to the idea of batch size [85], this enlarged input size facilitates faster convergence of the proposed CTDI model. As a result, for each missing data mask, the optimal number of epochs for CTDI to converge is fewer for the “AU” dataset than for the “A” dataset.

D. Discussion of CTDI Performance

The experimental results in Sections V-A–V-C lead to the following conclusions: 1) The proposed CTDI approach outperforms all the baseline models – including statistical imputations, shallow machine learning-based imputations, traditional deep learning-based imputations, and our previously proposed state-of-the-art approaches – regardless of the pattern of missing data (whether generated or taken from the real-world); 2) The recovery accuracy of CTDI decreases as the rate of missing data increases. All data recovery models are adversely affected by fewer observations from which to recover the missing data, since the information content is reduced. Additionally, a higher rate of missing data tends to reflect a higher proportion of consecutively missing data and previous work has shown that recovery of consecutively missing data is more difficult than that of sporadic missing data [13], [37]. The relative outperformance of CTDI as missing data rates increase and recovery becomes more challenging also suggests that CTDI improves on existing methods.

We attribute CTDI's superior performance to its transformer architecture, especially the Self-Attention mechanism. To explore this, Figs. 11–14 show the internal correlations of the air pollution data, calculated by aggregating and averaging the raw air pollution measurements into one 24-hour period and then using the Pearson correlation coefficient [86] to calculate the similarities between the averaged data. The value of this correlation coefficient is in the range of [0, 1] and a larger coefficient value indicates a higher correlation.

TABLE VI
CTDI ACCURACY FOR RECOVERING MISSING DATA AT DIFFERENT TIMES FOR CASE 3 AND CASE4

MAPE (%) for Different Air Pollutants												
Missing rate	00:00	01:00	02:00	03:00	04:00	05:00	06:00	07:00	08:00	09:00	10:00	11:00
2.46%	12.77	12.14	12.20	10.34	10.29	8.88	N/A	N/A	N/A	N/A	9.07	11.64
	12:00	13:00	14:00	15:00	16:00	17:00	18:00	19:00	20:00	21:00	22:00	23:00
	11.68	10.95	8.86	9.90	10.19	9.93	10.07	N/A	N/A	N/A	N/A	N/A
Missing rate	00:00	01:00	02:00	03:00	04:00	05:00	06:00	07:00	08:00	09:00	10:00	11:00
24.60%	11.71	17.51	13.91	13.01	13.14	12.17	10.77	11.04	10.53	10.80	11.27	12.13
	12:00	13:00	14:00	15:00	16:00	17:00	18:00	19:00	20:00	21:00	22:00	23:00
	12.90	12.72	12.42	11.52	11.76	11.93	11.13	11.07	11.25	10.80	10.50	10.86

TABLE VII
DATA RECOVERY COMPARISON FOR CASE 5

MAPE (%)										
Missing rate	Mean	MICE	KNN	MissForest	MIDA	LSTM	GINN	SVT	GSTR	CTDI
2.29%	16.05	15.12	15.92	15.41	14.77	15.00	14.36	14.94	11.58	10.71
5.47%	19.45	18.54	18.69	18.74	17.96	18.31	17.05	16.42	13.31	12.35
12.08%	28.91	24.66	24.64	24.45	21.58	20.96	20.07	22.51	15.72	14.44
23.13%	40.07	34.04	35.32	32.41	24.96	23.47	24.12	33.28	21.26	18.87

TABLE VIII
DATA RECOVERY COMPARISON BETWEEN AU AND A

MAPE (%)				
Missing rate	AU	Best epoch	A	Best epoch
2%	13.29	96	12.99	92
4%	13.91	77	13.16	97
6%	13.59	80	13.24	97
8%	13.88	81	13.48	58
10%	14.06	65	13.33	74
20%	14.45	30	14.18	35
30%	15.05	41	15.22	30
40%	15.90	32	15.85	42
50%	17.26	31	16.35	40
60%	18.57	22	18.55	24
2.46%	10.90	85	10.63	98
24.6%	13.05	94	12.75	98
2.29%	10.71	43	11.02	88
5.47%	12.35	54	12.11	67
12.08%	14.44	48	14.07	65
23.13%	18.87	75	18.52	85

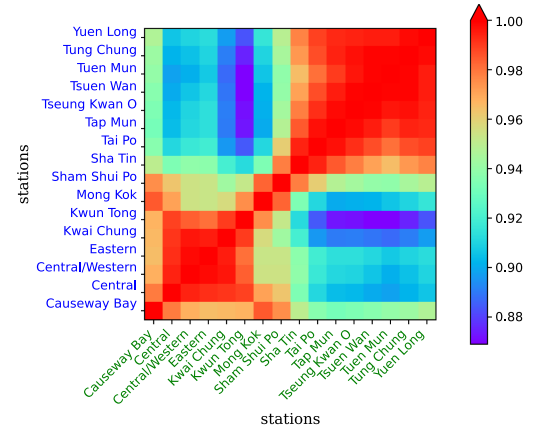


Fig. 12. Data correlation from different stations in the average daily air pollution data.

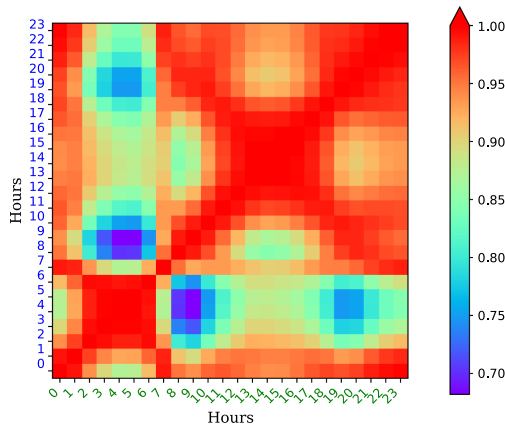


Fig. 11. Hourly data correlation in the average daily air pollution data.



Fig. 13. HK air pollution data monitoring stations.

Fig. 11 presents the correlation between air pollutant levels for different hours of the day. The correlations between commuting hours in the morning and afternoon and the correlations between early morning sleeping hours are extremely high (high

correlations are shown in red). Since the Self-Attention mechanism weighs the importance of all the positions within the input sequence and allows the model to update its weightings based on all relevant positions, irrespective of the distance

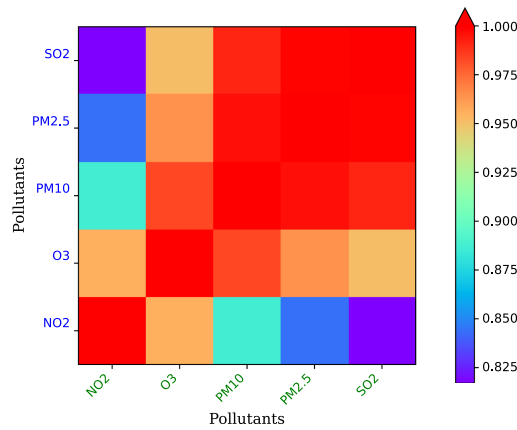


Fig. 14. Data correlation from different pollutants in the average daily air pollution data.

between the target position and those positions relevant to it, the Transformer-based CTDI model is able to better capture the temporal relationships present in time-series air quality-related data than CNN, which convolutes only on local neighbors, and RNN, which finds it difficult to capture long-term dependencies.

Similarly, Fig. 12 shows the correlation between different monitoring stations. The geographic locations of these stations are illustrated in Fig. 13. Monitoring stations that are located close to each other (such as Mong Kok and Sham Shui Po) have high correlations. Additionally, monitoring stations that are situated relatively far apart may still have high correlations, for example, city centers (such as Causeway Bay, Central, and Mong Kok), and lower activity areas (such as Sha Tin, Tap Mun, and Yuen Long); we hypothesize this is because their environments share similar characteristics, such as usage (shaping human activity and, in turn, pollution generation) and physical characteristics, such as street canyons and building density (shaping pollution propagation and concentration/dispersion). Through its attention mechanism, the Transformer-based CTDI model is able to capture spatial relationships in the inputs between adjacent and non-adjacent stations much more effectively than CNN, GCN, and RNN.

Finally, Fig. 14 shows the correlation between the different air pollutants. The 1-by-1 CNN layer allows the CTDI model to capture the relationships between different types of data features and hence improve accuracy by using information from other pollutants to recover the missing data of a particular pollutant. The combination of these features enables CTDI to consistently outperform other approaches, regardless of the particular pattern of missing data to be recovered.

VI. CONCLUSION AND FUTURE WORK

This study proposes a CNN-Transformer-based data imputation model named CTDI for the recovery of missing air pollution data. The model consists of three components. First, within the data pre-processing component, IDW is utilized to spatially interpolate air pollution data and related urban factors. A sliding window strategy is then employed to temporal segment the

data; this augments the dataset to provide the model with more 24-hour period data tensors from which to learn.

Second, a 1-by-1 convolution layer is employed after the data pre-processing component. This layer transforms of the depth of the 24-hour period data tensor and enables the model to capture the relationship and interactions between different types of data features. At the same time, this 1×1 CNN layer preserves the height and width of the data, ensuring the retention of the inherent spatial and temporal relationships between data entries situated in the S-T positions within the 24-hour period data tensor.

Third, the S-T Transformer module is specifically designed to capture the spatial and temporal relationships within the 24-hour period data tensor. The module achieves this effectively by transposing the input tensor into spatial and temporal sequences and then utilizing the powerful feature extraction capability of the Multi-Head Attention architecture. Additionally, the loss function of CTDI penalizes both entry-wise imputation error and overall average error to maximize the recovery of the reconstructed matrix.

Finally, we have also investigated the statistical characteristics of a Hong Kong air pollution dataset collected over 3 years. We found that the missing data rate across the full-time period is 2.46% and that the missing data is non-uniformly distributed. In particular, there is pronounced periodicity, with most missing data entries occurring in the early morning and noon periods. According to the investigation results, we then used a variety of patterns of missing data to evaluate CTDI. Simulation results show that CTDI outperforms baselines (including statistical imputation, shallow machine learning-based methods, conventional deep learning techniques, and two advanced recovery approaches we previously proposed.) in all scenarios. In addition, experiments on different datasets indicated that the air pollution observations alone are sufficient for the recovery of air pollution measurements in the case of Hong Kong. Furthermore, our ablation experiments indicate that each component makes a significant contribution to overall performance of CTDI, particularly highlighting the importance of the temporal transformer in scenarios with varying rates of missing data.

One direction of future research is to investigate the applicability and performance of CTDI in recovering air pollution data in other cities. In cities where meteorology data and traffic data are measured at the same sites as pollution monitoring stations, incorporating these urban factors may play a more significant role in the imputation of air pollutants than in the Hong Kong case.

In addition to air pollution data collected from air quality monitoring stations, the proposed CTDI model can also be directly applied to impute measurements from mobile air pollution sensors. The data from these sensors also exhibit complex spatial-temporal correlations and possess internal feature relevance, making them suitable for CTDI-based imputation. Moreover, we believe this could potentially be extended to other systems with similar characteristics, such as power systems and traffic flow, further expanding the applicability and usefulness of the CTDI model.

ACKNOWLEDGMENT

The authors gratefully acknowledge the Environmental Protection Department of the Hong Kong SAR Government, for providing valuable online open data source. The authors would also like to thank Dr. Yang Han, Department of Electrical and Electronic Engineering, the University of Hong Kong (HKU), for providing the urban dataset he has downloaded. All errors rest entirely on the authors.

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